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Short Communication

Viability of grid-connected solar PV energy system in Jos, Nigeria

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ABSTRACT

This study examines the feasibility of solar PV-grid tied energy system for electricity generation in a selected location in the northern part of Nigeria using HOMER energy optimization software. The technical and economic performance of a combination of 80 kW solar PV-grid connected was investigated. At base case of solar PV cost of \$2400/kW and average global solar radiation of 6.0 kW h/m^2/day, it was found that this energy system can generates annual electricity of 331,536 kW h with solar PV contributing 40.4% and the levelized cost of energy is \$0.103/kW h. Based on the findings from this study, the development of grid-connected solar PV system in the north-eastern part of Nigeria could be economically viable. The effects of the cost of PV system and global solar radiation were also investigated.

1. Introduction

Availability and utilization of energy is essential for social and economic development of a society and it is an essential resource required to improve human standard of living and quality of life. In Nigeria, access to reliable and stable supply of electricity is a major challenge for both the urban and rural dwellers. However, this problem is more significant in the rural areas and communities where only about 10% of the population have to access to electricity [1]. Even in the urban areas where grid-connected electricity is available, access to electricity is still a big challenge due to low and inadequate generation and distribution capacity. At the time of preparing this article, the peak electricity generation capacity in Nigeria is 3119.4 MW; which is about 24.4% of the peak electricity demand forecast of 12,800 MW for the same period [2]. This situation can be improved upon by using renewable energy resources, especially solar energy, to supplement the grid electricity supply in Nigeria. However, due to intermittent nature of these resources, they may not be suitable and reliable as stand-alone energy systems. Therefore, integration of both renewable energy conversion systems with storage facility could be a reliable energy system option in many locations in Nigeria.

In remote areas with no grid access, battery bank can be used as the storage facility. The negative effect of this is that additional cost of battery could significantly increase the unit cost of the electricity produced [3]. But, in areas with grid system, energy storage facility can be removed, and instead, the grid system can be used as 'storage' system. In this arrangement, when the renewable energy conversion system (RECS) produce more energy than needed, the surplus energy is fed into the grid and, energy is taken from the grid when the RECS system produces less energy than needed. As outlined by Mondal and Islam [4], some of the other advantages of PV-grid tied energy system includes: it can reduce energy and capacity losses in the utility distribution network, and it also can avoid or delay upgrades to the transmission and distribution network where the average daily output of the PV system corresponds with the utility's peak demand period. This arrangement is a good option for a PV-grid energy system in a tropical region like Nigeria, due to high availability of solar radiation in the country (see Section 2 below).

Feasibility, reliability and economic analyses conducted in a number of studies showed that hybrid power systems either as standalone or grid-tied system, are more reliable and cheaper than single source energy systems [see e.g., 5,6] and could produces less greenhouse gases when compare with fossil-fuel resources based energy systems [see e.g., 5–7]. The objective of this work is to investigate the techno-economic viability of solar PV-grid connected energy system in a location north-east Nigeria. This energy system may not only improve access to reliable supply of electricity, but can also reduce dependency on diesel generator systems (which are commonly used to supplement grid supplied electricity

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in semi-urban and urban areas across the country), and thereby reducing the associated noise pollution and emissions from these diesel generators. The energy system software, HOMER (Hybrid Optimization Model for Electric Renewable) is used to model the energy system and access its technical and economic performance.

2. Selected site and solar energy resource

Nigeria is located in western Africa on the Gulf of Guinea and has a total area of 923,768 km^2 and lies between latitudes 4° and 14°N, and longitudes 2° and 15°E. The solar radiation distribution in Nigeria is shown in Fig. 1. Three distinct different solar radiation zones can be identified and they are labeled as I, II and III with each zone having different range of solar radiation. The solar radiation in each zone can roughly be group as: zone I: 5.7–6.5 kW/m^2/day with sunshine hours of about 6 per day, zone II: 5.0–5.7 kW/m^2/day with sunshine hours of about 5.5 per day and, zone III: 3.5–5 kW/m^2/day with sunshine hours of about 5 per day.

For this study, a site (Jos, in Plateau state) located in the zone I is selected. This site is located on latitude 9°52′N and longitude 8°54′E and at an elevation of about 1238 m above sea level. The performance of a solar PV-grid-connected system is strongly dependent on the solar radiation which is site-specific and for a given site, its values vary frequently. Based on the data obtained from NASA Surface meteorology and solar energy website (NASA), the monthly averaged daily global solar radiation data for Jos is shown in Fig. 2. As expected, monthly variation in global solar radiation is observed and hence, the monthly energy output from solar PV would vary from one month to another. The monthly clearness index, which is defined as the fraction of solar radiation at the top of the atmosphere that reaches a particular location on the earth surface varied between 0.48 in the month of August (rainy season) and 0.70 in the month of December (dry season) with an annual average of 0.61. The prevailing weather condition in Jos can generally be considered as partly overcast weather (but close to clear weather condition around November and December).

3. Electrical load and energy system components

The daily electrical load used in this study is taken from the work of Ogbonna et al. [10]. They reported in detail the domestic energy consumption patterns in Jos, northern Nigeria. From a typical daily electricity consumption profile for this location (see Fig. 3), two prominent peak demand periods can be observed in daily electricity load profile from this figure and they occur in the morning, between 06.00 and 09.00 and; late in the evening, between 19.00 and 21.00. These electrical load peaks are due to usual morning activities (e.g., cooking of breakfast, lighting), and cooking of supper, lighting, TV, reading (in the evening/night).

They also noted that there is an insignificant difference between the daily electricity demand patterns for weekdays and weekends and the daily average demand is reported to be about 1 kW h. However, the simulation analysis carried out in this study is based on assumed electrical load of 750 kW h/day for an area with reliable electricity grid access. At an average consumption rate of 2.5 kW h per day, 300 households will be benefit from this installation. For this load profile, hourly and daily variations are taken as 15% and 25% respectively.

The proposed energy system comprises of solar energy conversion system (PV) and national electricity grid system. The schematic diagram of the solar PV and grid-connected system is shown in Fig. 4. Detailed descriptions of each component with the required input data are presented in the following sections.

3.1. Grid system

The grid purchase capacity and sales are taken as 100 kW and 60 kW, respectively and interconnection charge is assumed to be \$500. In this study, it was assumed that the consumers will buy certain amount of electricity from the grid at a price specified by the distribution company. The excess electricity generated will be sold back to the grid at 75% of grid price. In Nigeria, the cost of electricity is fixed and regulated by the Nigerian Electricity Regulatory Commission (NERC). The monthly electricity bill is

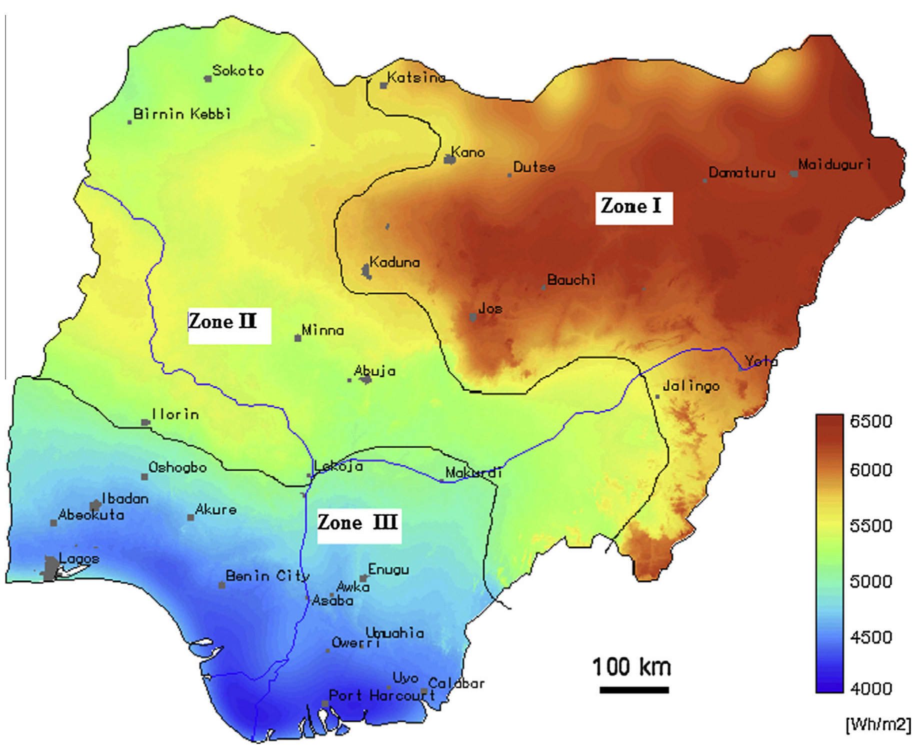

Fig. 1. Solar radiation map of Nigeria [8,9].

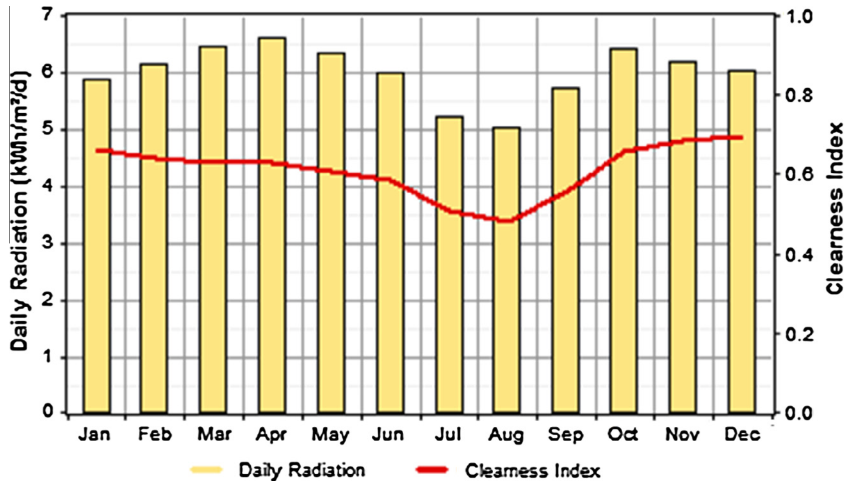

Fig. 2. Monthly daily averaged global solar radiation and clearness index for Jos.

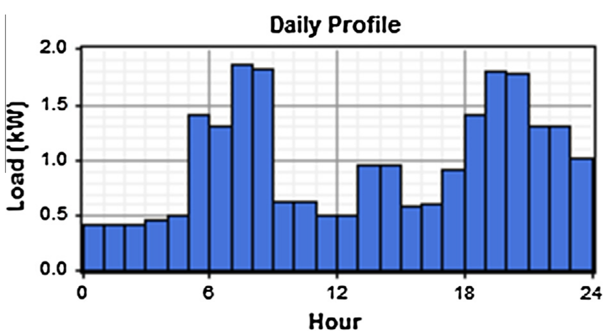

Fig. 3. Typical daily electricity consumption in Jos Nigeria [5,10].

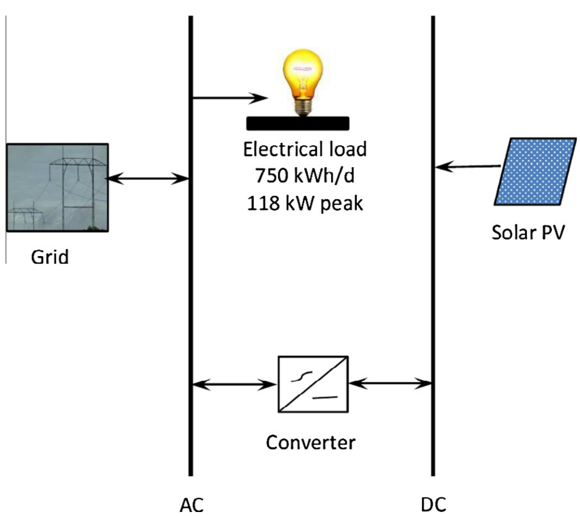

Fig. 4. The schematic diagram of the solar PV and grid-connected system.

made-up of fixed charge and energy consumed charge. Both of these charges vary across the country. The current monthly (2014) energy charge and fixed charge by distribution company in Jos area for residential users (*R*2) are N14.32/kW h and N1163, respectively and for residential users (*R*3), energy charge and fixed charge are N25.80/kW h and N43625, respectively [5]. The *R*2 and *R*3 users generally consumed more than 50 kW h of electricity per month and have the voltage connection systems of single and 3-phase, and LV (Low Voltage) Maximum demand, respectively.

Table 1
Technical specification of MaxPower CS6X-300P under nominal operating cell temperature [13].

Parameter		Specification
Nominal maximum power		300 W_p
Optimum operating voltage		32.9 V
Optimum operating current		6.61 A
Open circuit voltage		41.0 V
Short circuit current		7.19 A
Efficiency		14.71%
Operating temperature		−40 °C~ +85 °C
Temperature coefficients:	P_{max}	−0.43%/°C
	V_{oc}	−0.34%/°C
	I_{sc}	0.065%/°C
Normal operating cell temperature		45 ± 2 °C
Dimension		1954 mm × 982 mm × 40 mm

3.2. PV array and converter

The solar PV-grid system consists of 80 kW PV module. The solar PV module is a 72-cell (6 × 12) poly-crystalline (model number CS6X-300M, manufactured by Canadian Solar) with a rated power of 300 W_p and can produced maximum 600 V DC. The detailed technical specifications of this model are presented in Table 1. The lifetime of the PV array is assumed to be 25 years (which is equivalent to worldwide warranty provided by the manufacturer). The initial cost of the PV is taken as US$2322 (or US$2400) per kW (Adaramola et al. [5]). The derating factor and the ground reflectance are taken as 80% and 20%, respectively and no tracking system is included in the PV system.

Since the power output from the solar PV module is in DC, power inverter system is required to converter the PV power output to AC power. The size of the converter is determined using Solectria String Sizer tool [11] and it was found that one PVI 85KW/PVI 85KW-PE inverter model can supply the required AC power output. This inverter model is designed for grid-tied electricity system. The technical specifications of this model are presented in Table 2. The cost of this inverter model is given as $28,250 [12]. The shipment, import duty and related costs are assumed to be 30% of the cost of the inverter. Therefore, the total initial cost of the inverter model is taken as US$36725 (or US$432/kW). The lifetime of a unit is taken as 15 years with an average efficiency of 90%.

3.3. Project life and economy

The lifetime of the project is taken to be 25 years. According to the available information, the current interest rate and inflation

Table 2
The technical specifications of the selected inverter [14].

Parameter		Specifications
DC Input	Absolute max voltage	600 VDC
	MMPT voltage range	300–500 VDC
	Max operating	264 A
AC Output	Nominal voltage	208, 240, 480 or 600 VAC, 3-Ph (3 wire standard, 4 wire option)
	Continuous power (VAC)	85 kW
	Continuous current (VAC)	236/205/102/82A
Others	Peak efficiency	96.6/96.5/97.0/96.9/97.5%
	Ambient temp. range (full power)	−40°F to +131°F

rate in Nigeria are 12% and 8% respectively [15] and from this these, the annual real interest rate is determined as 3.7% using Fisher expression. This value represents current real interest rate. The initial fixed capital cost which can be used to prepare the site for the system and other initial installation costs is taken as $25000 and the overall system operation and maintenance cost of $1500 per year is assumed.

3.4. HOMER software

The energy system is designed and analyzed using HOMER software. Among the available softwares, HOMER is the most widely used optimization software for hybrid systems [16,17]. This is due to its many possible combination of renewable energy systems and ability to perform optimization and sensitivity analysis which makes it easier and faster to evaluate the many possible system configurations [16]. There are many studies that have used HOMER to examine the technical and economic feasibility of hybrid energy systems worldwide [4–7,18–23]. Detailed description of this software can be found in [24].

The output power of a PV array can be calculated from the following equation and the PV specifications [23].

$$P_{PV} = Y_{PV} f_{PV} \left(\frac{\overline{G_T}}{G_{T,STC}} \right) \left[1 + \alpha_p (T_c - T_{c,STC}) \right], \tag{1}$$

where P_{PV} is the rated capacity of the PV array, that is, the power output under standard test conditions in kW; f_{PV} is the PV derating factor [%], $\overline{G_T}$ is the solar radiation incident on the PV array in the current time step [kW/m^2], ($G_{T,STC}$ is the incident radiation at standard test conditions [1 kW/m^2], α_p is the temperature coefficient of power [%/°C], T_c is the PV cell temperature in the current time step [°C] and $T_{c,STC}$ is the PV cell temperature under standard test conditions [25 °C]. In a case where the effect of temperature on the PV array performance is neglected, α_p can assumed to be zero and Eq. (1) reduces to:

$$P_{PV} = Y_{PV} f_{PV} \left(\frac{\overline{G_T}}{G_{T,STC}} \right). \tag{2}$$

4. Results and discussion

4.1. Electricity generation

The electricity generated by the PV-grid energy system and the corresponding electricity consumed by the users (when the cost of PV is $2400/kW and global solar radiation is 6 kW h/m^2/day) is shown in Table 3. This table shows the total electricity produced by the energy system is 331,536 kW h/year which comprises of 133,867 kW h/year (40.4%) from the solar PV and 197,668 kW h/year (59.6%) from the grid. The generated electricity is utilized as follows: 82.6% (273,695 kW h/year) is consumed by the users, 13.4% (44,454 kW h/year) is sold back to the grid and 4.0% accounted for transmission losses.

Table 3
Electricity generated by the solar PV-grid system and end-use consumption pattern.

	Production			Consumption	
	kW h/yr	%		kW h/yr	%
Solar PV	133,867	40.4	AC load	273,695	82.6
Grid purchases	197,669	59.6	Grid sales	44,454	13.4
			Losses	13,387	4.0
Total	331,536	100.0	Total	331,536	100.0

Table 4
Performance parameters for the PV system.

Parameters	Quantity
Rated capacity (kW)	80
Mean output (kW)	15.3
Mean output (kW h/day)	367
Capacity factor (%)	19.1
Total production (kW h/year)	133,867

The performance parameters of the solar PV are presented in Table 4. The capacity factor, which is defined as the ratio of an actual energy output during a given period to the energy output that would have been generated if the system is operated at full capacity for the entire period, is given as 19.1%. This relatively low value is due to facts that in this study, the orientation of the system fixed tilt (at latitude). The capacity factor can be increased using tracking system. This factor is a useful parameter for users, developer and manufacturer of the solar PV energy system. It determines the economic viability of this energy system. The value of capacity factor for the PV determined in this study is comparable to findings from similar studies (see e.g., [25]). It can further be observed that the average daily energy output from the solar PV system is observed to be 367 kW h/day or 15.29 kW.

The monthly average of electricity produced by each component of the PV-connected energy system is presented in Fig. 5. The monthly and seasonal variations in amount of electricity produced by PV system and contribution from the grid can be observed from this figure. This is due to the variability in monthly global solar radiation. The maximum average monthly power generated by the solar PV is about 16.9 kW (in November) and the minimum of about 12.2 kW occurred in August. The monthly energy from purchased and sold to the grid is presented in Table 5. It can be observed from this table that the quantity of electricity purchased varies from 14,941 kW h in November when the PV produced highest amount electricity to 19,486 kW h in August when the PV produced the least amount of electricity. Similar variations in the electricity sold to the grid can also be observed from this table.

4.2. Economic analysis

The economic analysis of the solar PV-grid connected system is assessed by the following indicators: the levelized cost of electricity (LCOE) and net present cost (NPC) of the system. The effects of the cost of the PV and global solar radiation on these indicators are also presented in this section. The HOMER software determine the net present cost of the system as the different between the present value of all the costs of installing and operating the system over its

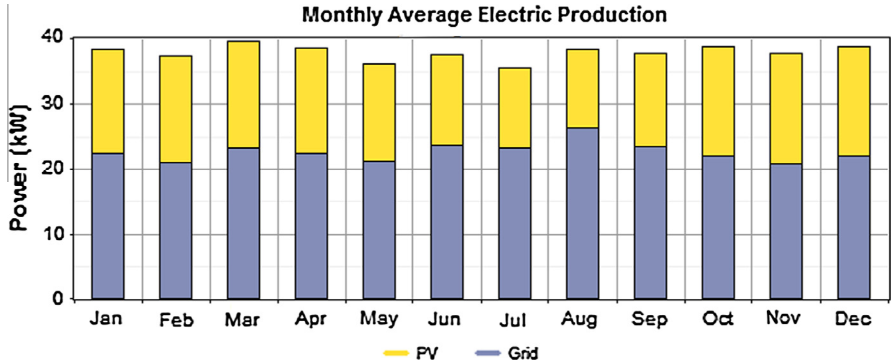

Fig. 5. Monthly distribution of the electricity produced by the energy system.

Table 5
Monthly amount of electricity purchased from and sold to the grid.

Month	Energy purchased (kW h)	Energy sold (kW h)
January	16,577	4503
February	14,089	4163
March	17,210	4154
April	16,104	3844
May	15,658	3645
June	16,914	3036
July	17,261	2480
August	19,486	2356
September	16,823	3202
October	16,327	4227
November	14,914	4404
December	16,306	4440
Annual	197,669	44454

Table 6
Effect of PV cost on the LCOE and NPC.

PV Cost (/$/kW)	Initial capital ($)	Operating cost ($/yr)	Total NPC ($)	LCOE ($/kW h)
2400	254,220	17,103	530,090	0.103
3000	302,220	17,103	578,090	0.113
3600	350,220	17,103	626,090	0.122
4200	398,220	17,103	674,090	0.131
4800	446,220	17,103	722,090	0.141
6000	542,220	17,103	818,090	0.159

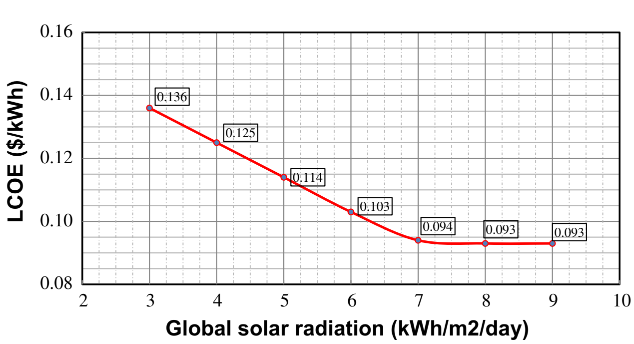

Fig. 6. Effect of global solar radiation on the LCOE when the PV cost is $2400/kW.

project lifetime, and the present value of all the revenues that it earns over the project lifetime. The variation of the NPC and LCOE with unit cost of the PV array is presented in Table 6. It should be mentioned that the presented LCOE and NPC are based on the net metering on monthly purchases from grid. For the base case; PV cost of $2400/kW, global solar radiation of 6.0 kW h/m^2/day and grid price of electricity of $0.090/kW h (for *R*2 residential users), the NPC for this project is $530,090, annual operating cost is $17,103 and the cost of the electricity is $0.103/kW h. As the cost of the PV increases, the LCOE, as expected, increases gradually from $0.103/kW h (when the PV is cost of $2400/kW) to $0.159/kW h (when the PV is cost of $6000/kW). The observed increment in the LCOE is primarily due to the increase in the initial capital cost of the PV which resulted in higher NPC when compared with the base case. It is expected that at lower PV cost (compared with the base case), the LCOE of the system will be reduced. For instance, when the simulation analysis was performed for PV cost of $1800 and $1200 per kW, the LCOE is found respectively as $0.081/kW h and $0.065/kW h. This shows that for the location considered in this study and similar locations in Zone I on the solar radiation map of Nigeria (see Fig. 1), solar PV-tied integration energy system is feasible and economically viable energy system.

The effect of global solar radiation on the LCOE is shown in Fig. 6 when the PV cost is $2400/kW. Two regions can be observed in this figure. These regions are: region 1 (when global solar radiation is less than or equal to about 7.0 kW h/m^2/day) - a linear relationship exists between the global solar radiation and the LCOE. In this region, the LCOE is observed to decrease with increasing global solar radiation. In region 2 (when global solar radiation is greater than about 7.0 kW h/m^2/day), the LCOE (at $0.093/kW h) is observed to be constant irrespective of the global solar radiation.

5. Conclusions

This study examines the feasibility of solar PV-grid tied energy system for electricity generation in a selected location in the northern part of Nigeria. The technical and economic performance of a combination of 80 kW solar PV and 100 kW power from the grid was investigated. It was found that this energy system can generates annual electricity of 331,536 kW h with solar PV contributing 40.4% and the levelized cost of energy is found to $0.103/kW h.

It is further observed that by reducing the initial installation costs (which consists of capital cost of the PV, connection cost and other associated costs), the cost of electricity can be significantly reduced. In addition, it was observed that the global solar radiation plays significantly impact on the economic viability of this system. It is expected that incorporating solar PV with grid system can reduce carbon dioxide and other pollutant emissions associated with thermal power plants generated electricity.

The information presented in this paper can serve as input to the development of grid-connected solar PV energy system in Nigeria. Based on the findings from this study, the development of grid-connected solar PV system in the north-eastern part of Nigeria could be economically viable energy system. However, as a result of high initial investment cost of solar PV system, favorable policies and incentives from government can accelerate the development of this type energy system in Nigeria. The logical next step from this study should be installation and performance assessment of practical grid-connected solar PV system in this location.

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